PROJECT-BASED LEARNING

PHASE-I EVALUATION

UNIGO

Team ID: T159

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T159
Semester: 3rd
Section: ML1, ML2, ML3, DS2
Domain: AI & ML, AI & DS

Team Members

1. Sapna Yadav – 2510011587
2. Srishti Nahar – 2510014554
3. Vanshika – 2510012685
4. Anushka Pun – 2512510037

Mentor

Ms. Megha Garg

Interactions completed: 3

Problem Statement & Motivation

Problem Statement

Our goal is to establish a centralized platform for finding campus locations, tracking bus routes, sharing feedback, and handling lost and found items.

Motivation

- *Information about the campus is spread across various sources.*
- *Searching or asking for help takes up time for both students and visitors.*
- *Existing methods rely on consulting others or through multiple sources.*

Target Users / Stakeholders

- *Students,faculty and visitors using the platform for campus assistance.*
- *administration, campus and transport staff manages information.*
- *Easy access to campus details helps save time and minimizes dependence on others.*

Objectives

Project Objectives

1. *Combine navigation, bus details, reviews and lost-found services in one platform.*
2. *C++ modules will handle graphs, hashing, searching, and other user data.*
3. *Data Structures handles navigation,and data management, while OOPS concepts for module organizaion.*
4. *Evaluate the core features using functional test cases.*
5. *The platform will tackle common campus issues and offer a flexible foundation for expanding campus facilities and services.*

Key Objective: *To minimize the time and effort needed to locate campus information and assist with common campus issues.*

Proposed Solution

Proposed Approach

1. *Official campus data, manual bus data ,lost-found from users.*
2. *Graphs for route finding,hashing or searching for quick data retrieval.*
3. *Find the request, retrieves data, processes through relevant module, and returns the result.*
4. *Structured records store campus, bus, reviews, and lost-found data.*
5. *The website will have interactive maps, routes, bus information, reviews, and lost-found details.*

Key Features

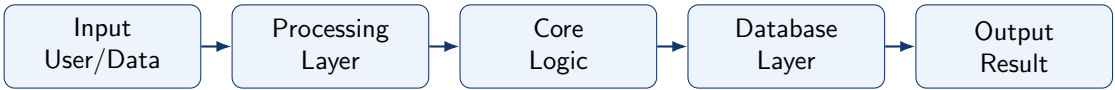
- *Interactive campus map to find locations and get suitable routes.*
- *Bus number,route,and timing information.*
- *Report lost or found items by providing description,images and location.*
- *Student reviews and feedback about campus locations and facilities.*
- *Access all major campus information and services through one focused platform.*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Graphs, Hashing, Searching, Queues, Arrays,etc.*	*Campus Navigation, Location Search modules*
OOPs with C++	*Classes, Objects, Encapsulation, Constructors, Inheritance, STL,etc.*	*Bus Routes modules*
Operating Systems	*Processes, Scheduling, Memory, Synchronization, etc.*	*Campus, Bus modules*
DBMS	*SQL, Transactions, Normalization, Indexing, etc.*	*Lost-Found,Review Modules*

Requirement: Each PBL subject will be meaningfully incorporated, with Data Structures managing data organization, searching, and navigation, whereas OOP principles will organize the various components of the projects.

System Architecture / Workflow



Major Components

- *Campus Navigation: Locates and finds routes on the campus map.*
- *Bus Information Module: Handles bus route management and schedules.*
- *Lost-Found and Review Module: Manages submitted user-recorded lost and found items and campus reviews.*

Data / Control Flow

- *User input → C++ backend → required data/DS processing → database → result.*
- *Modules-Navigation,Bus,Lost-Found,and Review,accesses their specific data via shared backend.*
- *Campus routes,bus details,Lost-Found records,and user comments are shown on the website.*

Technology Stack & Methodology

Technology Stack

- Programming: *C /C++/ HTML/ CSS/JavaScript*
- Database: *MySQL / PostgreSQL*
- Tools / Framework: *XXXX*
- IDE: *VS Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ and data structures enable basic operations and avigation, whereas web technologies deliver an accessible platform, and MySQL oversees the project's data.

Initial Progress & Team Contribution

Progress Achieved

- *Studied campus related common issues and lost-found items.*
- *Four core features finalized and data for modules was identified.*
- *workflow created linking the website, C++ backend, data structures, database.*
- *Navigation of campus,location-search module is the first development module.*
- *Field for campus data and other's needed for database preparation were identified.*

Individual Contribution

Member	Role	Contribution
M1	*Lead*	*Backend and DS*
M2	*Developer*	*Frontend*
M3	*DB/Testing*	*DB and Testing*
M4	*Developer/Docs*	*API and Docs*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Practical web platform having common required facilities.*
- *Less time-effort required to access campus information.*
- *Single platform to manage the common needs of students.*
- *Features can be expanded without affecting the basic system structure.*

References

1. *Faculty-provided notes and refrence books*
2. *GeeksforGeeks – Data Structures and C++*
3. *Google Maps Platform Documentation – Maps-location concepts.*
4. *Our GitHub Repository-Project source code annd data.*

Thank You
Questions & Suggestions